

ResuMetrics

Project Summary & Architecture Overview

What is ResuMetrics?

ResuMetrics is a smart assistant for hiring managers. Instead of spending hours reading through hundreds of resumes, you simply upload them to ResuMetrics. The system uses Artificial Intelligence to instantly read every resume, understand the candidate's experience, and rank them based on exactly what you are looking for in a perfect hire.

Key Features (Simply Explained)

- **Smart Reading (AI Understanding):** Instead of just using simple 'CTRL+F' to look for exact words, the AI actually understands meaning. If you ask for 'React', it knows that 'Frontend Web Developer' is a related skill.
- **Automatic Data Extraction:** The system automatically reads the resume and pulls out the important stuff: University names, Degrees, Job Titles, and impressive numbers (like 'Increased sales by 30%').
- **AI Candidate Summaries:** You don't have to read the whole resume. The AI writes a short, easy-to-read summary of the candidate's background, highlights their best soft skills, and even tells you if they get promoted quickly.
- **One-Click Emails:** Found some great candidates? You can select them all and hit 'Send Emails'. The system will automatically draft and send them personalized interview invitations.
- **Custom Take-Home Tests:** The AI can look at a candidate's specific skills and instantly generate a unique take-home assignment to test their abilities before the interview.
- **Visual Dashboard:** See all your candidates on a beautiful screen. You get visual charts (like a radar graph) showing exactly where they are strong and where they are weak.
- **Secure Storage:** Every resume and analysis is saved safely in a secure database so you never lose track of a good candidate.

Technology Used (Behind the Scenes)

- **The Brain (Python & Flask):** This is the main engine running the software, handling everything from uploading files to talking to the AI.
- **The Memory (MySQL Database):** A highly reliable storage system where all candidate information is kept safe.
- **The AI Reader (Sentence-Transformers):** A powerful AI model (similar to the tech behind ChatGPT) that understands human sentences.
- **The Look and Feel (HTML, CSS, JavaScript):** The code that makes the website look beautiful, modern, and easy to click around.
- **The Visuals (Chart.js):** The tool used to draw the colorful, interactive graphs on your screen.

How the Files are Organized

- **app.py:** The 'Traffic Controller'. It manages the web pages and connects the user to the database.
- **resume_parser.py:** The 'Heavy Lifter'. This is where the AI actually reads the resumes and calculates the scores.
- **static/js/main.js:** The 'Interactivity Engine'. This makes buttons work, charts draw, and animations play without reloading the page.
- **templates/:** The folder holding the visual layouts for the login screen and the main dashboard.